# The Manufacturing Clock of a Self-Replicating Fleet Is a Rounding Error on Galactic Settlement

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***Abstract*—A programme of self-replicating space manufacturing is usually modelled one scale at a time: a single factory that copies itself, a local fleet that disperses across a solar system, or a settlement front that crosses a galaxy. We thread one sourced factory through all three folds, so that a quantity every scale needs, the time to build one copy, is derived once rather than assumed three times over. The result is a clean cross-scale statement. The copy time that sets a local fleet's doubling clock, about 582 days for a lunar-regolith seed, is the same quantity that, at galactic scale, is under one part in a million of the roughly two-million-year time a powered front takes to fill a stellar field. The constraint that governs one scale is a rounding error at the next. Because that copy time is a lunar-factory stand-in for an interstellar probe whose own bill of materials is not yet known, we do not defend its exact value; we show the conclusion survives not knowing it. Across a copy-time sweep of five orders of magnitude the manufacturing dwell stays below one percent of the fill until the copy time is about fifteen thousand times its nominal value, so an error of four orders of magnitude in the build cadence does not change the verdict. Measured directly, seed by seed, the dwell costs a median of about a third of a percent of exploration time for the fastest slingshot policy and is within seed-to-seed noise for the slower one. The picture is therefore robust: manufacturing time is the binding clock for a local fleet, where transit is days, and a negligible tax on interstellar exploration, where transit is millennia.**

***Index Terms*—self-replication, interstellar exploration, cross-scale modelling, in-situ manufacturing, von Neumann probes**

## I. Introduction

A self-replicating machine that builds copies of itself from local material has been a fixture of long-range space-systems thinking since von Neumann's self-reproducing automata [1] and Freitas' self-reproducing interstellar probe [2]. Its appeal is leverage: a seed that remakes most of its own mass in place turns each launched kilogram into far more installed capacity than a finished payload could [3]. The same leverage, applied recursively, is what makes a single probe a plausible way to cross a galaxy in a time short against its age, the observation that sharpens the Fermi question [4], [5].

That programme is naturally studied at three very different scales, and in practice each is modelled on its own. At the smallest scale a *single factory* lands, closes on as much of its bill of materials as it can, and copies itself. At an intermediate scale a *local fleet* of such factories disperses across a solar system, each child building the next. At the largest scale a *settlement front* of self-replicating probes crosses interstellar space, and the exploration timescale is set by how the front chooses targets and how fast the probes travel [6], [7].

Modelling the three separately leaves a seam. Every scale needs one shared number, the time to build a single copy, and each has been free to pick its own. In the settlement-front model that number appears as a per-star *manufacturing dwell*: the time a freshly settled probe spends building offspring before they depart. For want of a sourced value it has been set to zero, so that the front is assumed to replicate instantaneously and only travel costs time. Whether that assumption is safe has never been checked against the factory physics that actually fixes the dwell.

This paper closes the seam and asks what the closed form says. We thread one sourced factory through all three folds and *derive* the manufacturing dwell from the very build physics the fleet already uses, so the number is computed once, from cited inputs, rather than assumed three times over. We introduce no new physical quantity; we route one the factory already fixes to the scale that was guessing it. The question we can then answer is cross-scale and quantitative: *which constraint binds at which scale?* The answer, developed below, is that the manufacturing cadence that is the binding clock for a local fleet is a negligible tax on interstellar exploration, and that this conclusion is robust to the large uncertainty in the one number it rests on.

## II. The cross-scale model

### A. One factory, three folds

Each scale is a separate, deterministic model that we drive from a single shared factory specification. The factory is the sourced lunar-regolith seed of the closure and replication literature [8]–[10]: a mass-closure figure $C$ (the fraction of its own mass it can remake locally) and a bill of materials whose masses and per-part energies are individually cited. We use it as a stand-in for a probe because a probe-specific bill of materials is not yet available in the literature (see Section IV); the point of this paper is that the choice does not matter to the conclusion.

The leverage that motivates the whole programme follows from mass balance alone: if a factory must import $(1-C)$ kg of parts for every kilogram it builds, then installed mass per launched mass is $1/(1-C)$, and the rocket equation makes shipping a finished installation instead exponentially more expensive in propellant [11]. Modern seed-factory studies place achievable closure in the range of roughly 70 to 96 percent and agree that chasing full closure is not worthwhile [3], [12], so a high but sub-unity $C$ is the realistic regime.

### B. The one derived number

The quantity every scale shares is the time to build one copy. At a heliocentric distance where the delivered solar power is known, a factory builds at a rate set by the lesser of its machinery throughput and the power its collectors can supply, and the time to lay down one offspring's worth of local structure is

$$t_{\mathrm{copy}} \;=\; \frac{C\, m_{\mathrm{seed}}}{\min(\dot{m}_{\mathrm{mach}},\, \dot{m}_{\mathrm{power}})}, \tag{1}$$

where $m_{\mathrm{seed}}$ is the seed mass, $\dot{m}_{\mathrm{mach}}$ the machinery-limited build rate, and $\dot{m}_{\mathrm{power}}$ the rate the available power supports at the per-kilogram embodied energy of the bill of materials [13]. We evaluate (1) at 1 AU, where the solar constant is 1361 $\mathrm{W\,m^{-2}}$ [14], consistent with the Sun-like uniform field the settlement-front model assumes. This is the same expression the fleet model already uses for its doubling cadence; nothing is added here.

The settlement-front model needs this number as a per-star manufacturing dwell in years,

$$\tau \;=\; t_{\mathrm{copy}}/Y, \qquad Y = 365.25\ \mathrm{d}, \tag{2}$$

where $Y$ is the Julian year ($1\ \mathrm{yr} = 3.15576\times10^{7}$ s). The same Julian year defines the speed of light in the units the front uses for light travel and stellar distances, so the dwell and the fill time it will be compared against are on one clock. A silent mismatch here would rescale the central ratio by orders of magnitude without any error elsewhere, so we hold it as a machine-checked invariant of the code.

### C. The finding, reduced to one ratio

With the dwell derived rather than zeroed, the front is run to completion and yields a fill time $T_{100}$, the time for the last star in the field to be settled. The entire cross-scale claim is the single dimensionless ratio

$$f \;=\; \tau/T_{100}, \tag{3}$$

one manufacturing dwell as a fraction of the whole galactic fill. If $f \ll 1$, the build cadence that governs the local fleet, where it is the only clock that matters because transit between neighbours is days, is a negligible tax at galactic scale, where transit is millennia. The rest of the paper measures $f$, tests how much it can be trusted given that $t_{\mathrm{copy}}$ is a proxy, and measures the dwell's cost on the front directly.

## III. Results

### A. The headline ratio

For the sourced lunar-regolith seed, (1) gives a copy time of about 582 days, so the derived dwell of (2) is $\tau \approx 1.59$ years. At fleet scale this same 582 days is the doubling clock: neighbours are days apart, so build time is the only clock that matters. Run as the settlement-front dwell for a powered cruise across a field of 1200 Sun-like stars, it sits against a fill time $T_{100} \approx 1.9 \times 10^{6}$ years, reproducing to the same order of magnitude the multi-million-year powered timescales of the model we build on [6]. The ratio of (3) is then

$$f \;\approx\; 8.5 \times 10^{-7}, \tag{4}$$

under one part in a million. The constraint that is the entire clock at fleet scale is, at galactic scale, a rounding error.

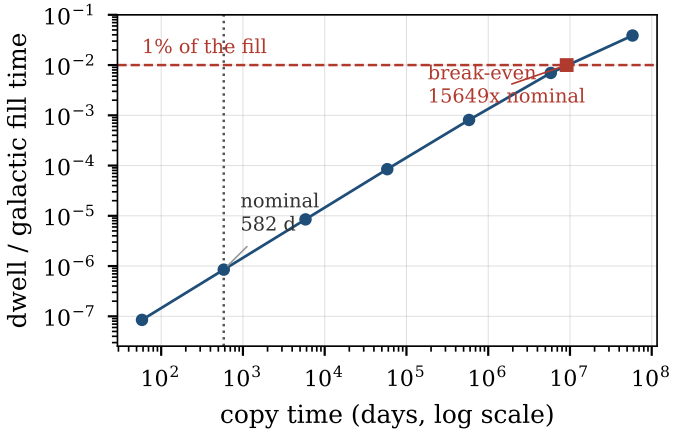



Fig. 1. The manufacturing dwell as a fraction of the powered galactic fill time, swept over five orders of magnitude in copy time (log-log; 1200-star field, event-resolved). The nominal lunar-seed copy time (582 days, dotted line) gives a fraction under $10^{-6}$. The fraction reaches the one-percent bar (dashed) only at a copy time about $1.5 \times 10^{4}$ times nominal (square), so an error of four orders of magnitude in the build cadence does not change the verdict.

### B. The result survives not knowing the copy time

The copy time is a lunar-factory proxy, so the case cannot rest on its exact value; it rests on the margin. Fig. 1 sweeps the copy time across five orders of magnitude and records the resulting dwell fraction of the powered fill. The fraction rises only in step with the copy time, and stays below one percent of the fill until the copy time is about $1.5\times10^{4}$ times its nominal value, that is, a copy time near 25 000 years rather than 582 days. A hypothetical probe whose true build cadence were four orders of magnitude slower than the lunar proxy would still leave the verdict unchanged. The conclusion is therefore insensitive to the one number we cannot yet source, which is the strongest statement we can make given the gap. The documented uncertainty in the collector inputs to (1), of order a factor of two, is a small sub-interval of this margin.

### C. The same holds across travel policies

The ratio is smallest for the slowest front and largest for the fastest, because a faster fill leaves the fixed dwell a larger share of a smaller total. Table I sweeps the three named policies on a common field and reports, for each, the break-even multiplier: the factor by which the copy time would have to grow before the dwell reached one percent of that policy's fill. Even for the fastest front, nearest-target slingshots, the copy time would have to be about sixty times larger, a probe taking a century rather than a year and a half to build one copy, before manufacturing became a one-percent effect. The

TABLE I

DWELL FRACTION AND BREAK-EVEN COPY-TIME MULTIPLIER BY TRAVEL POLICY, ON A COMMON 400-STAR FIELD (EVENT-RESOLVED). THE BREAK-EVEN MULTIPLIER IS THE FACTOR BY WHICH THE COPY TIME MUST GROW BEFORE THE DWELL REACHES ONE PERCENT OF THAT POLICY'S FILL TIME.

| Policy | Fill time (yr) | Dwell fraction | Break-even (× nominal) |
|---|---|---|---|
| Powered | $1.4 \times 10^{6}$ | $1.2 \times 10^{-6}$ | ∼10 000 |
| Slingshot max-boost | $4.8 \times 10^{4}$ | $3.3 \times 10^{-5}$ | ∼590 |
| Slingshot nearest | $8.3 \times 10^{3}$ | $1.9 \times 10^{-4}$ | ∼60 |

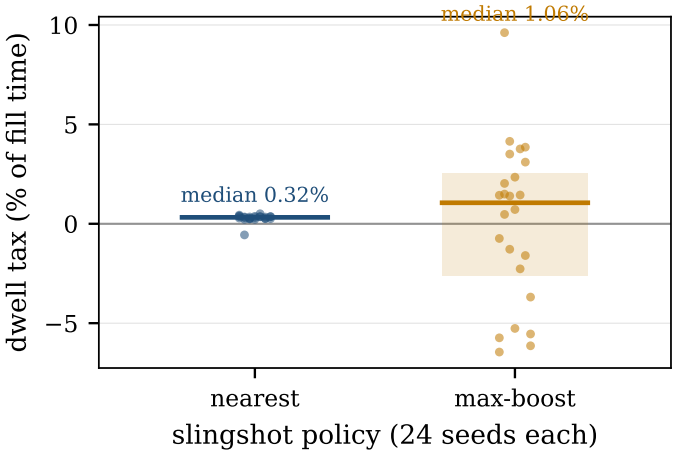



Fig. 2. The measured dwell tax (fractional increase in fill time from switching the derived dwell on) per seed for the two slingshot policies, 24 seeds each, event-resolved on a 400-star field. Bars are the ensemble median, shaded bands the interquartile range. Nearest-target slingshots give a tight, resolved 0.32 percent; max-boost scatters across zero and is within seed noise.

fastest speeds contemplated in the literature, beamed directed-energy cruise at 0.1 to 0.2 $c$ [15], lie beyond the slingshot regime we simulate and would push the crossover nearer still; the break-even multipliers bound how much nearer, and none of the simulated policies comes within a factor of sixty of it.

### D. Measuring the tax directly

The ratio $f$ is an analytic separation; we also measure the dwell's cost on the front directly, as the fractional increase in fill time when the derived dwell is switched on against the old zero-dwell baseline on the same field. For a multi-million-year powered fill this difference is far too small to resolve by simulation, which is itself the finding, so we measure it for the fast slingshot policies, where it is largest, over a 24-seed ensemble (Fig. 2). For nearest-target slingshots the tax is tight and clearly resolved at a median of 0.32 percent of exploration time (interquartile range 0.26 to 0.36 percent). For max-boost slingshots the per-seed values scatter from about $-6$ to $+10$ percent around a median near one percent: at this field size the dwell's effect is within the seed-to-seed noise of the fill time, so we report it as positive-but-unresolved rather than as a point estimate. Reporting a single seed here, as an earlier draft of the underlying finding did, would have hidden that spread.

Two checks confirm the measurement is not an artifact. Halving the integration timestep leaves the nearest-slingshot tax stable and converging on the event-resolved value, whereas a timestep coarse against the dwell under-resolves it, the same discretization hazard identified for the underlying front model; we therefore report event-resolved values throughout. Varying the field size over 200, 400, and 800 stars moves the median tax only between 0.31 and 0.34 percent, so the small-field measurement is representative over the range tested.

## IV. DISCUSSION

### A. Limitations

The load-bearing limitation is the numerator. The copy time of (1) is computed from a lunar-regolith seed because no per-module mass and energy breakdown yet exists for a concrete interstellar-probe design; the most developed such design reports its architecture but not a closed bill of materials at the fidelity this calculation needs [16]. We therefore do not present 582 days as a probe's copy time. The paper's whole defense against this gap is the margin of Fig. 1: the conclusion holds for any copy time within four orders of magnitude of the proxy, which comfortably spans the range a heavier, less-closed, or more power-starved probe could plausibly occupy. Closing the gap with a sourced probe bill of materials would replace the proxy with a measurement, but on the present evidence it cannot change the verdict.

The dwell is evaluated at 1 AU around a Sun-like star, matching the uniform field the settlement-front model assumes but not the real spread of stellar luminosities and material availability. Because stellar variation enters only through the numerator, its effect is already bounded by the same margin; a fainter or resource-poorer star lengthens the copy time, and it would have to do so by four orders of magnitude to matter. We deliberately do not add a stellar-population sub-model: it would be a separate slice with its own validation, not a parameter of this one, and the margin makes it unnecessary for the present claim. Likewise we inherit the front model's idealizations, chiefly a frozen stellar field, but our claim is a ratio in which the dwell enters the same way whether or not those idealizations hold, so it is insulated from them in a way an absolute fill time would not be.

Finally, the direct tax for max-boost slingshots is unresolved at the field size we can afford (Fig. 2); we report it as positive but within seed noise rather than as a number. That honesty is the point of the ensemble. The powered tax is smaller still and is knowable only analytically, through (3), not by brute force.

### B. Implications

The substantive implication is a division of labour between scales. Manufacturing cadence is the binding constraint on a local, densely packed fleet, where it sets the doubling time outright, and is a negligible tax on a sparse interstellar front, where propulsion policy and raw transit time dominate the timeline by six orders of magnitude. A programme that worries about how fast its seeds can build is asking the right question

for the solar-system phase and the wrong one for the galactic phase, where the same effort is better spent on propulsion. The result also illustrates a modelling stance: by keeping the three scales as separate, honestly sourced models joined at a clean seam, rather than one fused simulation, we can derive a shared quantity in the place it is physically fixed and route it to the place that needs it, and read off which constraint binds where. The seam is what makes the cross-scale statement possible; fusing the scales would have buried it.

## V. Conclusion

Threading one sourced factory through the single-factory, local-fleet, and galactic-front models lets a single quantity, the time to build one copy, be derived where the physics fixes it rather than assumed separately at each scale. Doing so answers a cross-scale question no one model can: the manufacturing cadence that is the entire clock for a local fleet, about 582 days for a lunar-regolith seed, is under one part in a million of a powered galactic fill, and the directly measured dwell tax on a fast slingshot front is a median of about a third of a percent of exploration time. Because the copy time is a proxy for a probe whose bill of materials is not yet known, the central result is stated as a robustness margin rather than a point value: the verdict holds for any copy time within four orders of magnitude of the proxy. Manufacturing time is thus the binding constraint on self-replication in a solar system and a rounding error on the way across a galaxy, where transit dominates. The natural next step is to replace the lunar-factory proxy with a sourced interstellar-probe bill of materials, which would turn the numerator from an estimate into a measurement without, on present evidence, moving the conclusion.

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